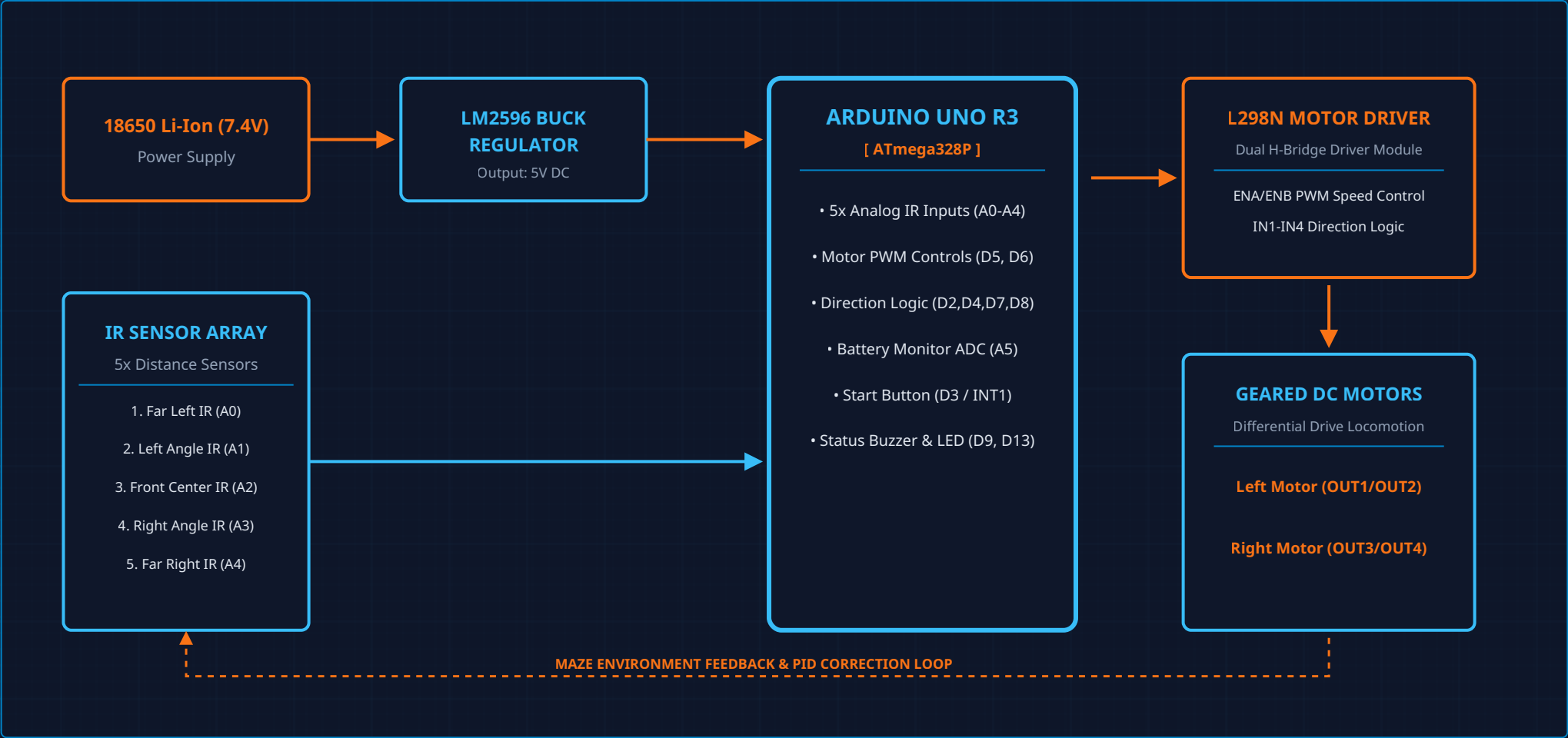


1. COMPLETE SYSTEM CIRCUIT OVERVIEW & BLOCK ARCHITECTURE



MICROMOUSE AUTONOMOUS ROBOT
BLOCK DIAGRAM & SYSTEM ARCHITECTURE

PROJECT LEAD: **VIKAS**
INSTITUTION: **IIT BOMBAY / IEEE**

DATE: **JULY 2026**
STATUS: **APPROVED FOR FABRICATION**

2. DETAILED WIRING INTERCONNECT & PIN ALLOCATION TABLE

Arduino UNO R3 Pin Connections

Arduino Pin	Target Component	Signal Function
A0	IR Sensor 1 (Far Left)	Wall Distance Analog Input
A1	IR Sensor 2 (Left Angle)	Wall Steering Alignment
A2	IR Sensor 3 (Front Center)	Obstacle & Dead-End Detection
A3	IR Sensor 4 (Right Angle)	Wall Steering Alignment
A4	IR Sensor 5 (Far Right)	Wall Distance Analog Input
A5	Battery Voltage Divider	7.4V Battery Monitoring
D2	L298N IN1	Left Motor Forward Control
D3	Start Push Button	User Trigger (Interrupt 1)
D4	L298N IN2	Left Motor Reverse Control
D5 (PWM)	L298N ENA	Left Motor Speed (PWM)
D6 (PWM)	L298N ENB	Right Motor Speed (PWM)
D7	L298N IN3	Right Motor Forward Control
D8	L298N IN4	Right Motor Reverse Control
D9	Piezo Buzzer	Acoustic Status Feedback
D13	Built-in LED	System Diagnostics Indicator

Power & Motor Driver Wiring Matrix

Source Component	Pin / Rail	Destination Pin
18650 Pack (+)	7.4V BAT (+)	SPST Toggle Switch Input
SPST Switch	Output	LM2596 IN+ & L298N 12V Pin
18650 Pack (-)	GND (-)	Common System Ground Rail
TP4056 Module	OUT+ / OUT-	Charging Port for 2S BMS
LM2596 Buck	OUT+ (5V DC)	Arduino 5V Pin & Sensor VCC
LM2596 Buck	OUT- (GND)	Arduino GND & Sensor GND
L298N Driver	OUT1 / OUT2	Left DC Motor Terminal A/B
L298N Driver	OUT3 / OUT4	Right DC Motor Terminal A/B
Voltage Divider	R1 (10k) / R2 (10k)	A5 Battery Sense Input Pin

Power Delivery Specifications

- Nominal Supply Voltage: 7.4V Li-ion (2S1P 18650 Configuration)
- Logic Regulation: LM2596 DC-DC Buck Converter (Step down to 5.0V @ 2A Max)
- Motor Driver Capacity: L298N Dual H-Bridge (Up to 2A per channel peak)
- Noise Decoupling: 100uF Electrolytic Capacitor on 5V Logic Rail + 0.1uF Ceramic caps across Motor Terminals

MICROMOUSE AUTONOMOUS ROBOT

INTERCONNECT SCHEMATIC & WIRING MATRIX

PROJECT LEAD: VIKAS
INSTITUTION: IIT BOMBAY / IEEE

DATE: JULY 2026
STATUS: APPROVED FOR FABRICATION

3. POWER DISTRIBUTION DIAGRAM & SENSOR ARRAY GEOMETRY

Power Distribution Topology

2S Li-Ion Battery

7.4V Nominal

5-Sensor Geometric Placement

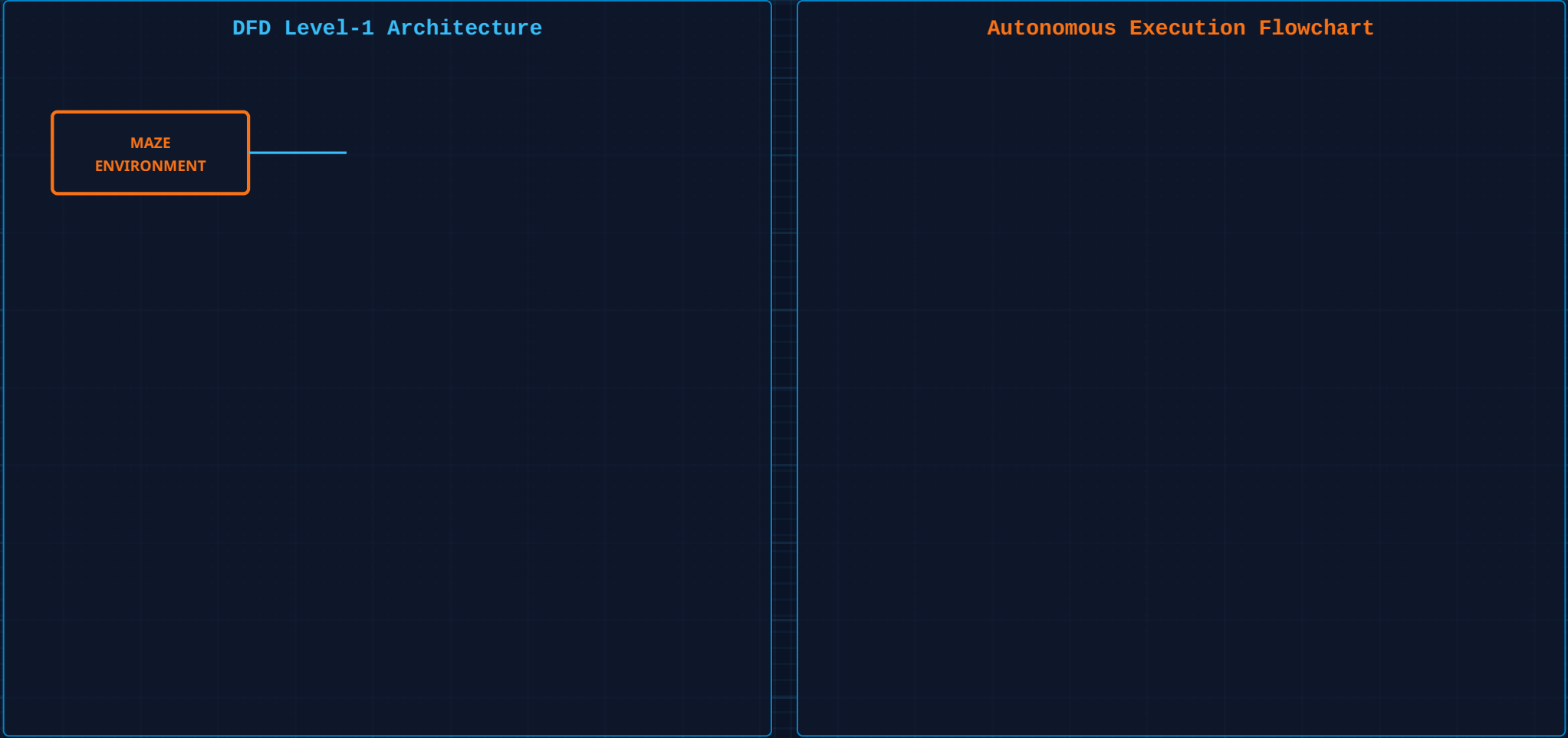
MICROMOUSE AUTONOMOUS ROBOT

POWER DISTRIBUTION & SENSOR GEOMETRY

PROJECT LEAD: VIKAS
INSTITUTION: IIT BOMBAY / IEEE

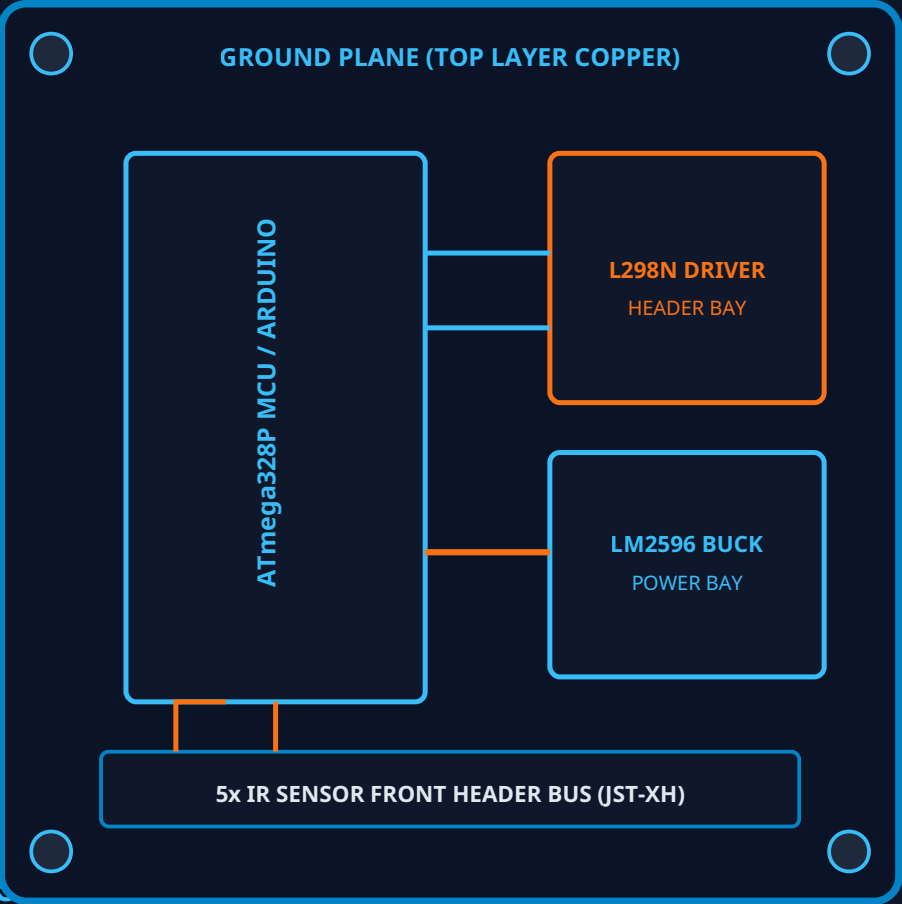
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4. SOFTWARE DATA FLOW DIAGRAM (DFD LEVEL-1) & EXECUTION FLOWCHART



5. CUSTOM PCB CONCEPTUAL DESIGN & PLACEMENT GUIDE

2-Layer Custom Printed Circuit Board (100mm x 100mm) Layout



MICROMOUSE AUTONOMOUS
CUSTOM PCB BLUEPRINT & ROUTING SPECIFICATION

INSTITUTION: IIT BOMBAY / IEEE

DATE: JULY 2026
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